Stockholm University

Department of Meteorology

Numerical Methods in Atmospheric Sciences and Oceanography (MO8007)

Lab 1 – The Advection Equation

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1 Introduction

Numerical weather models are an essential tool for modern society. They help predict the weather and climate, which is crucial for traffic and trade, societal safety related to weather, and predicting climate change. These models work by discretizing equations on a spatial–time grid, by applying different numerical schemes. In this report, we will study two common schemes used in meteorology, oceanography, and atmospheric physics: the Upwind scheme and the Leapfrog scheme. This will be done with the information and schemes presented in Döös et al. [2022].

When dealing with numerical models there are four main types of computational errors that can affect the result. These include: When dealing with numerical models there are four main types of computational errors that can affect the result. These include:

1. Truncation error
2. Stability error
3. Computational mode
4. Phase speed error.

The truncation error is the most fundamental error, which arises from the finite differences method, for approximating derivatives, gives other non-zero Taylor coefficients which are neglected. The stability error is often a larger error if not treated correctly. It arises when three time steps are used in a scheme that results in there being two theoretical solutions to the scheme. These two roots are labelled the physical and computational mode, where the computational mode is an error arising from the discretization while the physical mode is the actual solution. However the output will always be a superposition of the two modes. The last major error arises from the fact that the phase speed of the analytic solution and the numeric scheme are not the same. This is due to the fact that the computational scheme uses gridded space-time, which will slightly alter the propagation speed.

1.1 The numerical models

The advection equation is defined as

$$\frac{\partial u}{\partial x} + c\frac{\partial u}{\partial t} = 0. \tag{1}$$

This can be solved analytically by observing that c is simply the propagation speed, from an initial condition $u(t = 0, x)$. However this can also be discretized using different schemes.

1.1.1 The leapfrog scheme

In the Leapfrog scheme we use finite differences as

$$\frac{u_j^{n+1} - u_j^{n-1}}{2\Delta t} + c\frac{u_{j+1}^n - u_{j-1}^n}{2\Delta x} = 0 \implies u_j^{n+1} = u_j^{n-1} - \mu\left(u_{j+1}^n - u_{j-1}^n\right), \tag{2}$$

where $x = j\Delta x$, $t = n\Delta t$ and $\mu = c\Delta t/\Delta x$ is the Courant number. Here the Truncation error is in the order of $\sim \mathcal{O}\left[(\Delta x)^2\right] \sim \mathcal{O}\left[(\Delta t)^2\right]$. Since this is a iterative method of finding the nex step, one has two start with two time-steps for this to work. The most common choice for this second time-step is to use the forward Euler method

$$\frac{u_j^{n+1} - u_j^n}{\Delta t} + c\frac{u_{j+1}^n - u_{j-1}^n}{2\Delta x} = 0 \implies u_j^{n+1} = u_j^n - \frac{\mu}{2}\left(u_{j+1}^n - u_{j-1}^n\right). \tag{3}$$

The Leapfrog scheme is stable if one chooses a Courant number $|\mu| \leq 1$, since this will always yield a stability parameter $\lambda = 1$.

1.1.2 The upwind scheme

The Upwind scheme uses the fact that we know which direction propagation comes from. If we know that the propagation is to the right ($c >$) gives

$$\frac{u_j^{n+1} - u_j^n}{\Delta t} + c\frac{u_j^n - u_{j-1}^n}{\Delta x} = 0 \implies u_j^{n+1} = u_j^n - \mu\left(u_j^n - u_{j-1}^n\right). \tag{4}$$

Here the Truncation error is higher at $\sim \mathcal{O}\left[\Delta x\right] \sim \mathcal{O}\left[\Delta t\right]$. The stability here is still $\mu < 1$. However, the stability parameter is dependent on μ, which leads to a Courant number less than one yielding a stability parameter less than one and thus diffusion of the amplitude with time.

1.1.3 The phase speed error

One interesting concept to keep in mind, when dealing with these numerical models is that the phase speed observed from the models may differ from the theoretical phase speed. If one looks at the Leapfrog scheme one can derive

$$\frac{c_D}{c} = \frac{1}{\mu k \Delta x} \arcsin\left[\mu \sin(k\Delta x)\right], \tag{5}$$

where c_D is the computational phase speed and c is the theoretical phase speed. One can thus observe a larger phase speed error when μ becomes smaller than one and when $k\Delta x$ decreases.

1.1.4 Robert-Asselin filtering

There are different ways to get rid of this computational mode and thus get cleaner results. One common way is to use some kind of filter, such as the Robert-Asselin filter. The Robert-Asselin filter works by smoothing the result, since the computational mode adds background noise. This is done as

$$u_{\text{filtered}}^n = u^n + \gamma\left(u_{\text{filtered}}^{n-1} - 2u^n + u^{n+1}\right), \tag{6}$$

where γ is a filtering parameter. The Robert-Asselin filter has the drawback in that a it also dampens the result by a factor $[1 - 4\gamma\sin(\omega\Delta t/2)]$ due to the dispersion-like smoothing. Furthermore, this parameter also affects the stability criteria so that, for example, in the Leapfrog scheme where the new stability condition becomes $\mu + \gamma \leq 1$.

2 Methodology

The goal of this project was to test the Leapfrog scheme (Equation 2) and the Upwind scheme (Equation 4) for a cosine wave and a cosine pulse. Thus, in the first part, an initial condition of the form

$$u(x, t = 0) = \cos\left(2\pi x\right) \tag{7}$$

was used. The two schemes were computed iteratively in Fortran using a spatial grid of $\Delta x = 0.02$ arbitrary length units over an extent $L = 1$ arbitrary length units, a theoretical phase speed of $c = 1$ arbitrary velocity units, for the Courant numbers $\mu = [1.0, 0.9, 1.1]$. This was firstly examined together with the analytic solution in a Hovmöller diagram for the time extent $t = 2.5$ arbitrary time units.

Later the schemes were allowed to run for $t = 10^4$ arbitrary time units to examine how the error changed with time. Firstly the phase and amplitude error were analysed. Secondly the relative error from the analytic solution was computed for the different schemes and Courant numbers.

In the second part a cosine pulse of the form

$$u(x, t = 0) = \begin{cases} \frac{1}{2} + \frac{1}{2}\cos\big(10\pi(x - 0.5)\big), & 0.4 \leq x \leq 0.6, \\ 0, & \text{elsewhere}, \end{cases} \tag{8}$$

was used. The same parameters were used in this part, except that only a Courant number of $\mu = 0.9$ was analysed. The models were left to run for $t = 2$ arbitrary time units. A Robert-Asselin filter was then applied to study the removal of the computational mode. A smoothing parameter of $\gamma = 0.1$ was chosen.

3 Results

3.1 The cosine wave

The results from the Leapfrog scheme and the Upwind scheme, for the different Courant numbers, can be seen in Figure 1. There, one can observe that a Courant number of $\mu = 1.0$ gave an identical result to the analytical solution, while a Courant number of $\mu = 1.1$ made both schemes become unstable. A Courant number of 0.9 gave slightly damped peaks for the Upwind scheme, while the Leapfrog scheme showed no noticeable deviation from the analytical solution at this timescale.

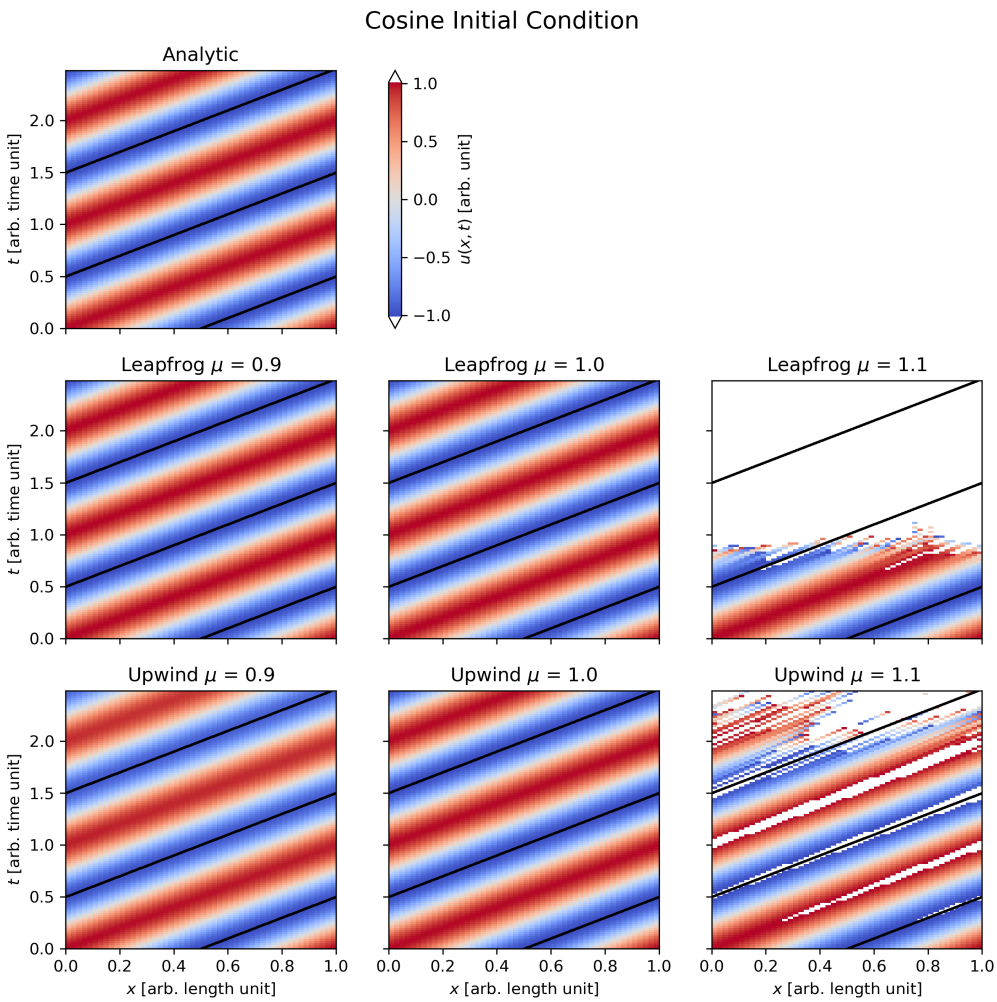

Figure 1: This figure shows the solution for the leapfrog scheme, the upwind scheme, along with the analytical solution for various Courant numbers. The black line shows the propagation speed of the analytic solution. Values above or below the analytical values are left as blanc.

In Figure 2, one can observe the phase and amplitude errors from the Leapfrog scheme and Upwind scheme. In both schemes, it can be seen that the numerical schemes are slightly slower than the analytical one. Comparing this value with the theoretical value in Equation 5, one obtains $c_D/c \approx 0.9995$, which

is in agreement with the observed phase shift in Figure 2. Interestingly, a large amplitude drop can be seen in the Upwind scheme, while this is not observed in the Leapfrog scheme.

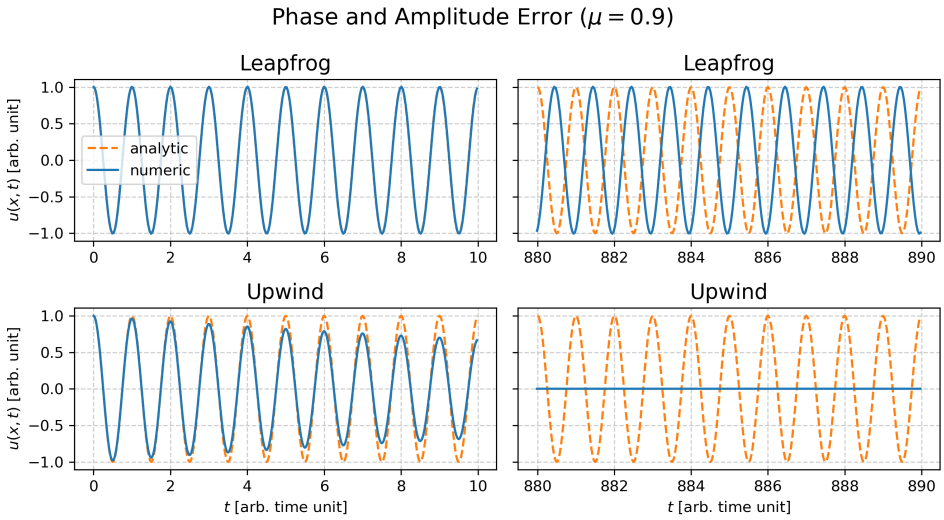

Figure 2: This figure shows the phase and amplitude errors for the Leapfrog scheme and Upwind scheme.

The relative error from the analytic solution can be seen for the different schemes and Courant numbers in Figure 3. There, one can observe that a Courant number of $\mu = 1.1$ yielded very large errors after approximately $t \approx 1.0$ arbitrary time units for the Leapfrog scheme and $t \approx 2.5$ arbitrary time units for the Upwind scheme. Furthermore, a Courant number of $\mu = 0.9$ for the Upwind scheme did not become unstable, although a non-negligible error arises that grows with time. Both $\mu = 0.9$ and $\mu = 1.0$ in the Leapfrog scheme show small, but present errors, while $\mu = 1.0$ in the Upwind scheme shows very small errors.

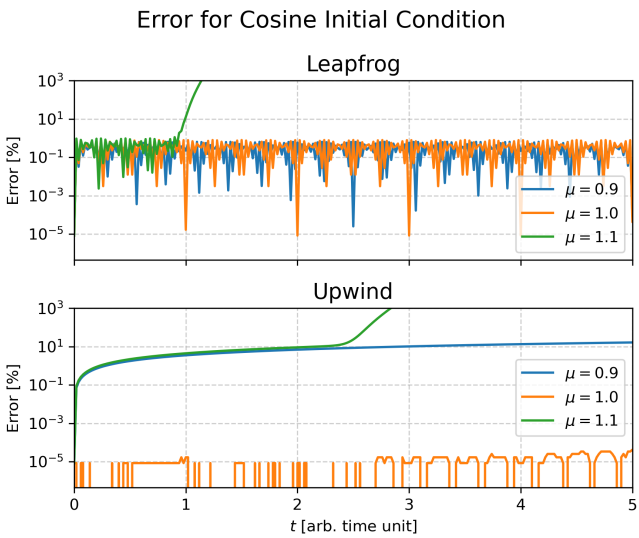

Figure 3: This figure shows the errors over time, for the different Courant numbers in the Leapfrog scheme and Upwind scheme.

3.2 The cosine pulse

In Figure 4, one can observe the analytic, Leapfrog, and Upwind results side by side for $\mu = 0.9$. Comparing them with the analytic solution, one can observe oscillations after the pulse for the Leapfrog

scheme that grow with time. Due to these being oscillations, one can assume this error to originate from the computational mode. In the Upwind scheme, no oscillations can be seen; however, temporal dissipation can be observed.

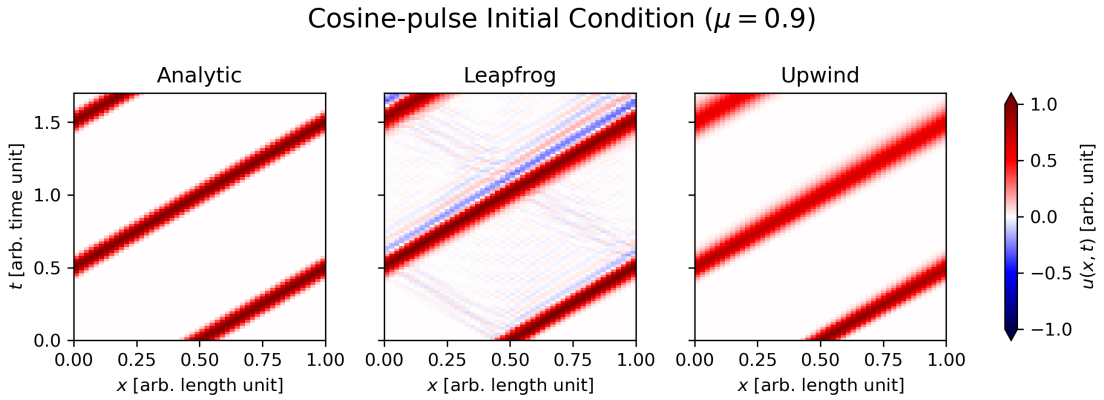

Figure 4: This figure shows the solution for the leapfrog scheme, the upwind scheme, along with the analytical solution.

In Figure 5, one can observe what a Robert-Asselin filter, for three different filter parameters, does to the Leapfrog scheme in Figure 4. One can observe that a filter parameter of $\gamma = 0.1$ removed the computational mode oscillations, although an amplitude decrease can be seen. Choosing $\gamma = 0.05$ also proved to remove the computational mode without diffusing the data excessively. However choosing $\gamma = 0.01$ proved not to remove the computational mode.

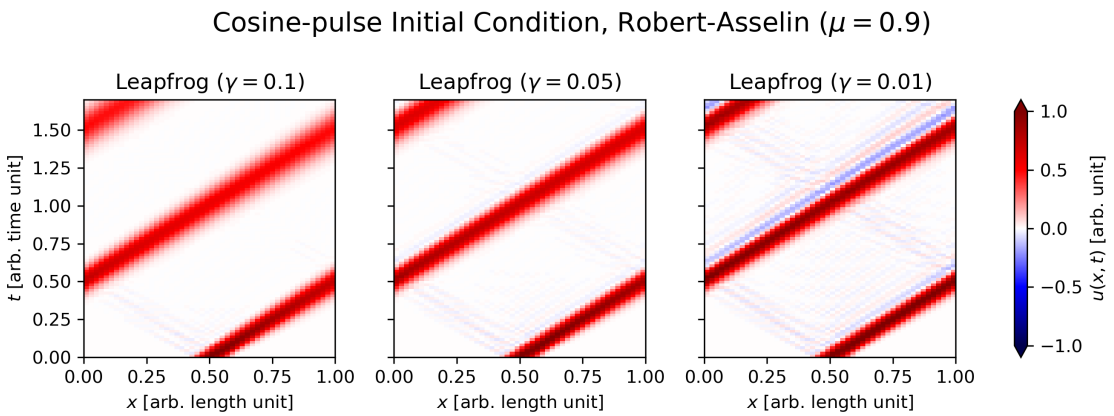

Figure 5: This figure shows the effect of the Robert-Asselin filters on the Leapfrog scheme.

4 Discussion

The result in Figure 1–5 is overall in agreement with the theory presented above. It was shown that both the Leapfrog and the Upwind scheme could be used to model the advection equation, if the Courant number was chosen $\mu \leq 1$. When $\mu = 1.1$, an unstable result was obtained; however, one must note that the Leapfrog scheme blew up before the Upwind scheme. Furthermore, when the Courant number was chosen as $\mu = 1$, the Upwind scheme showed the smallest error. However, for $\mu = 0.9$, a smaller error could be seen in the Leapfrog scheme. Furthermore, a large amplitude diffusion could be seen in the Upwind scheme, which was expected from the theory of the scheme. However the Leapfrog scheme showcased no such error. Wherever the phase error was the same for the Upwind scheme becomes irrelevant since the amplitude error is so large.

The reason for the Upwind scheme showing larger errors for $\mu \neq 1$ might be due to the truncation error being larger in the Upwind scheme. Thus, one would expect larger deviations from the analytical solution in the Upwind scheme. One can then question why the Upwind scheme showed a much smaller error when the Courant number was chosen as $\mu = 1$. However, one has to remember that the Leapfrog scheme is initialized with the forward Euler scheme. Thus, the initial conditions for the Leapfrog scheme may have an error from the forward Euler scheme, which might explain the larger error for the $\mu = 1$ case compared to the Upwind scheme.

Although it was clear that a Courant number $\mu = 1$ gave the cleanest result, one should note that this parameter will be hard to use in more difficult equations. More advanced equations will give larger computational modes, which will lead to one being reliant on filtering the data. Since one needs a non-zero γ-parameter, one cannot choose $\mu = 1$. It is therefore more interesting to study what $\mu = 0.9$ does to the system.

A Courant number $\mu = 0.9$ proved to give stable, reliable results for the cosine wave, both for the Leapfrog scheme and the Upwind scheme. For the cosine pulse, a Courant number $\mu = 0.9$ gave oscillatory results for the Leapfrog scheme and diffused results for the Upwind scheme. However, introducing the Robert-Asselin filter (with $\gamma > 0.01$) proved to remove the computational mode for the Leapfrog scheme. To reduce diffusion from the Robert-Asselin filter, a filter parameter of $\gamma = 0.05$ gave the cleanest result.

5 Conclusion

In this report, the advection equation was studied through the Leapfrog scheme and the Upwind scheme. These schemes were studied for different Courant numbers and compared with the analytical solution. A Courant number of $\mu = 1.1$ proved to give unstable results from both schemes, while a Courant number of $\mu = 1$ gave almost identical results to the analytical solution. However, due to the nature of more complex equations, a Courant number of exactly $\mu = 1$ is often unrealistic. Therefore, the Courant number of $\mu = 0.9$ is more interesting to study. For a cosine wave, $\mu = 1$ proved to give very good results for both schemes, although the Upwind scheme showcased diffusive properties over time. Both schemes showcase a slower phase speed than the theoretical one, although this was a very slight error. For a cosine pulse, the Courant number of $\mu = 0.9$ gave diffusive properties for the Upwind scheme and a visible computational mode for the Leapfrog scheme. However, applying a Robert-Asselin parameter to the Leapfrog scheme proved to remove this computational mode.

Overall, this lab has showcased the powerful tools of numerical models through the Leapfrog and Upwind schemes. The advection equation is an equation that can be solved analytically, but the results in this report showcase that these schemes can be used to study even more difficult equations and give reasonable results.

References

Kristofer Döös, Peter Lundberg, and Aitor Aldama Campino. *Basic Numerical Methods in Meteorology and Oceanography.* Stockholm University Press, Stockholm, Sweden, 2022. ISBN 978-91-7635-172-7. doi: 10.16993/bbs.

A Acknowledgements

The text in this report was spell-checked using Mistral AI's Le Chat. The schemes in this report were obtained using the Fortan programming language, while the figures were created using the Python programming language.